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Rotary electric machine

Abstract

A rotary electric machine includes a multi-phase winding including magnetic poles circumferentially arranged in a circumferential direction. The multi-phase winding includes crossover wires. Each crossover wire connects two single coils functioning as two of the magnetic poles that are same in phase. The crossover wire includes an inter-pole crossover wire and an end crossover wire. The inter-pole crossover wire extends between one end and another end of the multi-phase winding in an axial direction of the multi-phase winding. The inter-pole crossover wire extends through an inter-pole gap between adjacent two of the magnetic poles. The end crossover wire extends in the circumferential direction on the other end of the multi-phase winding.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2007/0182265	12/2006	Makino et al.	N/A	N/A
2017/0018985	12/2016	Suzuki	N/A	H02K 3/522
2019/0097475	12/2018	Sekii	N/A	H02K 1/2791
2021/0384790	12/2020	Echizen et al.	N/A	N/A

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) The present application is a continuation application of International Patent Application No. PCT/JP2020/022630 filed on Jun. 9, 2020, which designated the U.S. The entire disclosure of the above application is incorporated herein by reference.

TECHNICAL FIELD

(1) The disclosure in this specification relates to a rotary electric machine.

BACKGROUND

(2) Crossover wires of a stator coil are stacked along an insulator of a stator. The disclosure of a prior art literature (JP 2007-236181 A corresponding to US 2007/0182265 A1) is incorporated herein by reference to explain technical elements presented herein.

SUMMARY

(3) According to at least one embodiment of the present disclosure, a rotary electric machine includes a multi-phase winding including magnetic poles circumferentially arranged in a circumferential direction. The multi-phase winding includes crossover wires. Each crossover wire connects two single coils functioning as two of the magnetic poles that are same in phase. The crossover wire includes an inter-pole crossover wire and an end crossover wire. The inter-pole crossover wire extends between one end and another end of the multi-phase winding in an axial direction of the multi-phase winding. The inter-pole crossover wire extends through an inter-pole gap between adjacent two of the magnetic poles. The end crossover wire extends in the circumferential direction on the other end of the multi-phase winding.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The details of one or more embodiments are set forth in the accompanying drawings and the description below. Other features and advantages will be apparent from the description and drawings, and from the claims.
- (2) FIG. 1 is a top view showing a rotary electric machine according to a first embodiment.
- (3) FIG. 2 is a cross-sectional view taken along a line II-II of FIG. 1.
- (4) FIG. 3 is a cross-sectional view showing a connection portion of a conductive member.
- (5) FIG. 4 is a winding diagram of a stator.
- (6) FIG. 5 is a top view showing a rotary electric machine according to a second embodiment.
- (7) FIG. 6 is a perspective view showing a rotary electric machine.
- (8) FIG. 7 is a cross-sectional view on a line VII-VII in FIG. 5.
- (9) FIG. 8 is a perspective view showing a stator.
- (10) FIG. 9 is a top view showing a terminal block excluding a stator core.
- (11) FIG. 10 is a side view in an arrow X in FIG. 9.
- (12) FIG. 11 is a side view in an arrow XI in FIG. 9.
- (13) FIG. 12 is a cross-sectional view taken along a line XII-XII of FIG. 9.
- (14) FIG. 13 is a winding diagram of a stator.
- (15) FIG. 14 is a winding diagram of a stator according to a third embodiment.
- (16) FIG. 15 is a winding diagram of a stator according to a fourth embodiment.
- (17) FIG. 16 is a perspective view of a stator according to a fifth embodiment.
- (18) FIG. 17 is a winding diagram of a stator according to a fifth embodiment.
- (19) FIG. 18 is a perspective view showing a stator according to a sixth embodiment.
- (20) FIG. 19 is a winding diagram of a stator according to a sixth embodiment.
- (21) FIG. 20 is a top view of a stator according to a seventh embodiment.
- (22) FIG. 21 is an enlarged top view showing a part of the stator.
- (23) FIG. 22 is a perspective view showing the stator.
- (24) FIG. 23 is an enlarged perspective view showing a part of the stator.
- (25) FIG. 24 is a cross-sectional view showing the stator.
- (26) FIG. 25 is an enlarged cross-sectional view showing a part of the stator.
- (27) FIG. 26 is an enlarged cross-sectional view showing a part of the stator.
- (28) FIG. 27 is an enlarged top view showing a part of the stator.
- (29) FIG. 28 is an enlarged perspective view showing a part of the stator.
- (30) FIG. 29 is an enlarged cross-sectional view showing a part of the stator.
- (31) FIG. 30 is an enlarged top view showing a part of a stator according to an eighth embodiment.
- (32) FIG. 31 is an enlarged perspective view showing a part of the stator.
- (33) FIG. 32 is an enlarged cross-sectional view showing a part of the stator.
- (34) FIG. 33 is an enlarged top view showing a part of the stator.
- (35) FIG. 34 is an enlarged perspective view showing a part of the stator.
- (36) FIG. 35 is an enlarged cross-sectional view showing a part of the stator.
- (37) FIG. 36 is an enlarged top view showing a part of a stator according to a ninth embodiment.
- (38) FIG. 37 is an enlarged perspective view showing a part of the stator.
- (39) FIG. 38 is an enlarged cross-sectional view showing a part of the stator.
- (40) FIG. 39 is an enlarged top view showing a part of the stator.
- (41) FIG. 40 is an enlarged perspective view showing a part of the stator.
- (42) FIG. 41 is an enlarged cross-sectional view showing a part of the stator.
- (43) FIG. 42 is an enlarged top view showing a part of a stator according to a tenth embodiment.
- (44) FIG. 43 is an enlarged perspective view showing a part of the stator.

- (45) FIG. **44** is an enlarged cross-sectional view showing a part of the stator.
- (46) FIG. **45** is an enlarged top view showing a part of a stator according to an eleventh embodiment.
- (47) FIG. **46** is an enlarged perspective view showing a part of the stator.
- (48) FIG. **47** is an enlarged cross-sectional view showing a part of the stator.
- (49) FIG. **48** is an enlarged top view showing a part of a stator according to a twelfth embodiment.
- (50) FIG. **49** is an enlarged perspective view showing a part of the stator.
- (51) FIG. **50** is an enlarged cross-sectional view showing a part of the stator.
- (52) FIG. **51** is a top view showing a rotary electric machine according to a thirteenth embodiment.

DETAILED DESCRIPTION

- (53) To begin with, examples of relevant techniques will be described. According to a comparative example, crossover wires of a stator coil are arranged in a stack along an insulator of a stator.
- (54) In the comparative example, a rotary electric machine having a short length in an axial direction may be required.
- (55) In contrast, the present disclosure can provide a compact rotary electric machine.
- (56) The present disclosure can provide a rotary electric machine easy to be manufactured.
- (57) The present disclosure can provide a rotary electric machine small in size in an axial direction.
- (58) According to an aspect of the present disclosure, a rotary electric machine provides a multi-phase winding with multiple magnetic poles arranged in the circumferential direction. The multi-phase winding includes crossover wires. Each crossover wire connects two single coils providing two of the multiple magnetic poles that are the same in phase in the rotary electric machine. Each crossover wire includes an inter-pole crossover wire and an end crossover wire. The inter-pole crossover wire extends between one end and the other end of the multi-phase winding in an axial direction of the multi-phase winding. The inter-pole crossover wire extends through an inter-pole gap between adjacent two of the multiple magnetic poles. The end crossover wire extends in the circumferential direction on the other end of the multi-phase winding.
- (59) According to the rotary electric machine, the crossover wire connecting the two single coils that are the same in phase includes the inter-pole crossover wire as a part of the crossover wire. The inter-pole crossover wire extends between the one end and the other end in the axial direction through the inter-pole gap which is between two magnetic poles adjacent to each other in the circumferential direction. Since the inter-pole crossover wire provides a part of the crossover wire, a length of the end crossover wire is reduced. As a result, the number of end crossover wires of different phases is reduced, and a compact rotary electric machine can be provided.
- (60) Hereinafter, embodiments will be described with reference to the drawings. In the embodiments, parts that functionally and/or structurally correspond to or are associated with each other may be assigned the same reference numeral, or reference numerals different in digit in the hundreds or higher place. With respect to the parts that correspond to or are associated with each other, explanations thereof can be shared among the embodiments.

First Embodiment

- (61) FIGS. **1** and **2** show a rotary electric machine **1**. In FIGS. **1** and **2**, the rotary electric machine **1** is illustrated in a slightly exaggerated manner in radial direction. Dash lines indicate omissions or hidden lines. The rotary electric machine **1** is configured to rotate around a rotation axis AX. In the following descriptions, a direction in which the rotation axis AX extends is called an axial direction, a circumference around the rotation axis AX is called a circumferential direction, and a radiation direction centered at the rotation axis AX is called a radial direction.
- (62) In FIG. **1**, the rotary electric machine **1** is coupled to a rotary body **2**. The rotary body **2** is configured to rotate around the rotation axis AX. The rotary body **2** is a rotary shaft or an input/output end of a transmission. The rotary electric machine **1** is housed in a housing **3**. The housing **3** provides a fixation portion for the rotating rotary electric machine **1**.
- (63) The rotary electric machine **1** has a rotor **10** and a stator **20**. The rotary electric machine **1** is an

inner rotor type. The rotor **10** and the stator **20** are arranged so as to form an air gap AG therebetween. The rotor **10** is disposed outward of the rotary body **2** in the radial direction. The stator **20** is disposed further outward of the rotor **10** in the radial direction.

(64) The rotor **10** includes a rotor core **11** and multiple rotor magnetic poles **12**. The rotor core **11** is an annular magnetic body. The rotor core **11** is connected to the rotary body **2** in the rotation direction. The multiple rotor magnetic poles **12** are provided by permanent magnets. The multiple rotor magnetic poles **12** are arranged at equal intervals on an outer circumferential surface of the rotor core **11**. As a result, the rotor **10** provides a permanent magnet rotor.

(65) The stator **20** includes a stator core **30**, an insulator **40** attached to the stator core **30**, and a coil **50** attached to the insulator **40**. The stator **20** provides multiple stator magnetic poles. In this embodiment, the stator **20** provides eighteen stator magnetic poles. In the drawings, a stator magnetic pole **21**, a stator magnetic pole **22**, and a stator magnetic pole **23** are illustrated for example. Each of these three stator magnetic poles **21**, **22**, **23** includes a corresponding phase winding. One stator magnetic pole includes one tooth **31**, one bobbin **41** and one single coil **51**.

(66) These multiple stator magnetic poles define an inter-pole gap PG between two stator magnetic poles that are next to each other in the circumferential direction. The inter-pole gap PG is also called a circumferential gap or an inter-coil gap. The multiple inter-pole gaps PG have equal predetermined widths in the circumferential direction. The multiple inter-pole gaps PG are disposed at equal intervals in the circumferential direction. The inter-pole gap PG has a size that allows a connection portion described later to be arranged therein. The inter-pole gap PG has a size that allows a manufacturing operation for forming and arranging the connection portion to be performed. The size of the inter-pole gap PG can secure electrical insulation necessary as the rotary electric machine **1** between the connection portion described later and the coil **50**.

(67) The stator core **30** is, for example, a laminated body of electrical steel sheets. The stator core **30** includes multiple teeth **31**. The stator core **30** includes a yoke **32**. The yoke **32** is an annular magnetic body. The yoke **32** connects the multiple teeth **31** magnetically and mechanically. The multiple teeth **31** are arranged at equal intervals on an inner circumferential surface of the yoke **32**. The multiple teeth **31** and the annular yoke **32** are a continuous body.

(68) The insulator **40** is made of electrically insulating resin. The insulator **40** is a resin molded body. The insulator **40** includes multiple divided bodies divided in the axial direction. The multiple divided bodies are attached to the stator core **30** such that the insulator **40** is provided. The insulator **40** provides multiple bobbins **41** for the multiple teeth **31**. One bobbin **41** is formed on one tooth **31**. The bobbin **41** provides a winding drum for the coil **50**. One bobbin **41** has a tubular portion **42** located radially outward of one tooth **31**. One bobbin **41** provides a base end flange **43** at a base end of the tooth **31** and a top end flange **44** at a top end of the tooth **31**. In other words, the tubular portion **42**, the base end flange **43**, and the top end flange **44** provide one bobbin **41**.

(69) The coil **50** provides a stator winding. The coil **50** provides a multi-phase winding. In this embodiment, the coil **50** provides a three phase winding. The insulator **40** is arranged between the coil **50** and the stator core **30**. The coil **50** is arranged radially outward of the multiple teeth **31**. The coil **50** includes multiple single coils **51**. One single coil **51** is arranged radially outward of one tooth **31**. The multiple single coils **51** are formed by winding wires around the bobbins **41**. The wires are single wires made of copper or copper alloy. The wires have flexibility to allow winding work.

(70) The stator magnetic pole **21** provides one phase winding of the three phase winding, for example, a U-phase winding. The stator magnetic pole **22** provides another phase winding of the three phase winding, for example, a V-phase winding. The stator magnetic pole **23** provides the remaining phase winding of the three phase winding, for example, a W-phase winding.

(71) The stator **20** includes at least one conductive member **60**. The stator **20** includes multiple conductive members **60**. The conductive member **60** is also called a bus bar. The conductive member **60** has a cross-sectional shape different from that of the wire forming the coil **50**. The wire

has a circular cross section, but the conductive member **60** has a rectangular or square cross section. The conductive member **60** has flexibility. The flexibility of the conductive member **60** is lower than the flexibility of the wire. The conductive member **60** is harder and less likely to be deformed than the wire. Therefore, the conductive member **60** is a conductive member for electrically pulling out the wire of the coil **50** from the single coil **51**.

(72) The multiple conductive members **60** include power end members **61**, **62**, **63** and a neutral point member **64**. The power end members **61**, **62**, **63** provide output terminals or input terminals as the three-phase winding. When the rotary electric machine **1** functions as a generator, the power end members **61**, **62**, **63** provide output terminals. When the rotary electric machine **1** functions as an electric motor, the power end members **61**, **62**, **63** provide input terminals. In this embodiment, the power end members **61**, **62**, **63** provide terminals of an electrical connector. The neutral point member **64** provides a neutral point connection as the three phase winding.

(73) One conductive member **60** has at least one connection portion **65**, **66**, **67**, **68**, **69**. The conductive member **60** is electrically and mechanically connected to at least one coil end **52** at the connection portion **65**, **66**, **67**, **68**, **69**. The coil end **52** is an end portion of the coil **50**. The stator **20** includes multiple coil ends **52**. If the coil **50** provides a three-phase winding, the stator **20** has, for example, six coil ends **52**.

(74) The multiple power end members **61**, **62**, **63**, respectively, have multiple connection portions **65**, **66**, **67** which are arranged, respectively, in three inter-pole gaps PG next to each other. The power end member **61** has a connection portion **65** arranged in a first inter-pole gap PG. The power end member **62** has a connection portion **66** arranged in a second inter-pole gap PG. The power end member **63** has a connection portion **67** arranged in a third inter-pole gap PG. The first to third inter-pole gaps PG are arranged next to each other. Each of the multiple connection portions **65**, **66** and **67** is connected to a smallest number of coil ends **52**. The smallest number is a number of a parallel number in the coil **50**. In this embodiment, the smallest number is 1. When the coil **50** is provided by two parallel coils, the smallest number is 2.

(75) The neutral point member **64** has multiple connection portions **68**, **69** which are arranged, respectively, in the multiple inter-pole gaps PG next to each other. In an example illustrated in the drawings, the neutral point member **64** has two connection portions **68**, **69** which are arranged, respectively, in two inter-pole gaps PG next to each other. The connection portion **68** is connected to the smallest number of coil ends **52**. The connection portion **69** is connected to twice the smallest number of coil ends **52**.

(76) The multiple connection portions **65**, **66**, **67**, **68**, **69** are dispersively arranged in the multiple inter-pole gaps PG next to each other. The multiple connection portions **65**, **66**, **67**, **68**, **69** are arranged in the multiple inter-pole gaps PG next to each other, in a one-to-one relationship therebetween. In this embodiment, one connection portion is arranged in one inter-pole gap PG. As a result, the five connection portions **65**, **66**, **67**, **68**, **69** are arranged in the five inter-pole gaps PG next to each other.

(77) The connection portions **65**, **66**, **67**, **68**, **69** are positioned in the inter-pole gaps PG. The connection portions **65**, **66**, **67**, **68**, **69** are positioned in the inter-pole gaps PG in the axial direction. A part of the conductive member **60** may extend out of the inter-pole gap PG in the axial direction. However, the connection portions **65**, **66**, **67**, **68**, **69** are entirely arranged in the inter-pole gaps PG. The connection portions **65**, **66**, **67**, **68**, **69** are positioned in the inter-pole gaps PG in the radial direction. A part of the conductive member **60** may extend out of the inter-pole gap PG in the radial direction. However, the connection portions **65**, **66**, **67**, **68**, **69** are entirely arranged in the inter-pole gaps PG.

(78) Multiple magnetic poles protrude in the radial direction from the yoke **32** that extends in the circumferential direction. The multiple conductive members **60** include a circumferential extending portion **60a** and a radial extending portion **60b**. The circumferential extending portion **60a** extends in the circumferential direction along the yoke **32**. The radial extending portion **60b** extends in the

radial direction from the circumferential extending portion, and an end of the radial extending portion **60b** reaches the inter-pole gap PG. The connection portions **65**, **66**, **67**, **68**, **69** are formed on the end of the radial extending portion **60b**. For example, the power end members **61**, **63** have the circumferential extending portion **60a** and the radial extending portion **60b**. The power end member **62** includes only the radial extending portion **60b**. The power end members **61**, **62**, **63** include a radial extending portion extending inward toward the inter-pole gap PG for the connection portion, and a radial extending portion extending outward to be connected to an external circuit for external connection. The neutral point member **64** includes a circumferential extending portion **60a** and two radial extending portions **60b**.

(79) The stator **20** includes a terminal block **80**. The terminal block **80** is made of an electrically insulating resin. The terminal block **80** supports the multiple conductive members **60**. The terminal block **80** supports the multiple power end members **61**, **62**, **63**. The multiple power end members **61**, **62**, **63** are insert-molded on the terminal block **80**. The terminal block **80** has a body portion **81** and a connector portion **82**. The body portion **81** extends in an arc shape along the stator **20**. The connector portion **82** is located outward of the body portion **81** in the radial direction and extends outward from the body portion **81** in the radial direction. The connector portion **82** is connected to a connector of an external circuit. The connector portion **82** provides a connection between the power end members **61**, **62**, **63** and the external circuit. The external circuit provides a control circuit for the rotary electric machine **1**. The terminal block **80** is fixed to the stator **20**. Specifically, the terminal block **80** is fixed to the insulator **40**. The neutral point member **64** is supported by the insulator **40**.

(80) FIG. 2 shows a cross section taken along line II-II in FIG. 1. In this drawing, the position of the power end member **61** in the multiple conductive members **60** is illustrated by a relative positional relationship with the multiple stator magnetic poles. The multiple conductive members **60** are arranged similarly to the illustrated one. The drawing illustrates a crossover wire **54**. The arrangement and the number of the crossover wires **54** are merely examples. Arrangement and the number of the multiple crossover wires **54** are shown in winding diagrams described later.

(81) The multiple magnetic poles **21**, **22**, **23** include the insulator **40** as the bobbins **41**. A height TH**40** of the insulator **40** in the axial direction of the rotary electric machine **1** defines a height of the stator **20**. In other words, the height TH**40** of the insulator **40** defines a height of the rotary electric machine **1**. The multiple conductive members **60** are arranged within the height TH**40** of the insulator **40** in the axial direction.

(82) The coil **50** includes the crossover wire **54** extending over the multiple single coils **51**. The crossover wire **54** connects the multiple single coils **51** belonging to one phase winding by a continuous wire. In other words, the crossover wire **54** connects the multiple stator magnetic poles in the same phase by the continuous wire. In the case of three-phase winding, for example, the multiple single coils **51** belonging to the same phase, such as No. 1, No. 4, No. 7 . . . , are connected. The crossover wire **54** is laid along the insulator **40**. The crossover wire **54** extends at least partially on an outer side of the base end flange **43** in the radial direction. The crossover wire **54** extends at least partially on an end portion in the axial direction which is opposite to an end portion in the axial direction where the terminal block **80** is arranged.

(83) The power end member **61** extends in the radial direction in the connector portion **82** for external connection. The power end member **61** extends along the circumferential direction on the outer side of the base end flange **43** in the radial direction. The power end member **61** has a corner on an outer side of the first inter-pole gap PG in the radial direction. The power end member **61** extends across the base end flange **43** in the radial direction on the outer side of the first inter-pole gap PG in the radial direction. The power end member **61** extends in the first inter-pole gap PG in the axial direction. The connection portion **65** is positioned in the first inter-pole gaps PG. The connection portions **65** is positioned at the substantially center of the inter-pole gaps PG in the axial direction. The connection portions **65** is positioned at the substantially center of the inter-pole gaps

PG in the radial direction.

(84) The terminal block **80** is located outward of the base end flange **43** in the radial direction. The terminal block **80** is arranged at only one of opposite end portions of the rotary electric machine **1** in the axial direction. The connector portion **82** is located outward of the base end flange **43** in the radial direction. The connector portion **82** is exposed to an outside of the housing **3**. The connector portion **82** has an opening facing outward in the radial direction in the outside of the housing **3**. The connector portion **82** forms an electrical connection via the power end member **61** by receiving a connector of the external circuit moved along the radial direction from the outside in the radial direction.

(85) FIG. **3** is an enlarged view of the connection portion in FIG. **2**. The multiple conductive members **60**, that is, the multiple power end members **61**, **62**, **63** and the neutral point member **64** have similar shapes in the connection portions **65**, **66**, **67**, **68**, **69** provided by them. The conductive member **60** has a curved portion **60c** and a joint portion **60d** in the radial extending portion **60b**. The curved portion **60c** is a crank type that extends from an end surface of the stator **20** in the axial direction along a surface of the insulator **40** and further extends toward the inter-pole gap PG. The connection between the conductive member **60** and the coil end **52** is provided by fusing. The joint portion **60d** that provides the connection by fusing is provided by the conductive member **60** being bent so as to wrap the coil end **52**. The conductive member **60** and the coil end **52** are electrically and mechanically connected by fusing processing.

(86) FIG. **4** is a winding diagram showing a circuit of the coil **50** in the rotary electric machine **1**. The rotary electric machine **1** provides a multi-phase winding with multiple magnetic poles arranged in the circumferential direction. The coil **50** is illustrated as viewed from an inside of the stator **20**. In the figure, slots S7 to S15 which are repeated parts are omitted. The coil **50** includes multiple phase windings. The coil **50** includes a U-phase winding **50u**, a V-phase winding **50v**, and a W-phase winding **50w**. The coil **50** is wound such that the single coil **51** of the U-phase winding **50u**, the single coil **51** of the V-phase winding **50v**, and the single coil **51** of the W-phase winding **50w** are alternately formed. In this embodiment, eighteen magnetic poles are formed, and eighteen slots S1 to S18 are formed between the multiple magnetic poles. The slots S1 to S18 correspond to the inter-pole gaps PG. The stator **20** has one end **20a** and another end **20b** in the axial direction. The one end **20a** is an end portion where the conductive member **60** is arranged. The other end **20b** is an end portion where an end crossover wire **57** is arranged.

(87) The coil **50** is wound from a predetermined first magnetic pole. When winding of the single coil **51** in the first magnetic pole is completed, the winding of the coil **50** is moved to the next magnetic pole of the same phase through the crossover wire **54** and performed on this magnetic pole. Therefore, the crossover wire **54** extends between the two single coils **51** that provide the same-phase magnetic poles. The crossover wire **54** is also called a jumper wire. The coil **50** is wound on the multiple magnetic poles in sequence. In the following descriptions, the first magnetic pole will be called No. 1. Due to this winding process, a winding start portion **55** and a winding finish portion **56** of one single coil **51** in one magnetic pole can be specified. The crossover wire **54** includes the end crossover wire **57**, an inter-pole crossover wire **58**, and an inter-pole crossover wire **59**.

(88) The end crossover wire **57** is arranged on the other end **20b** of the stator **20**, i.e. the rotary electric machine **1**. In other words, the end crossover wire **57** is arranged on the end (the other end **20b**) opposite from the end (the one end **20a**) where the conductive member **60** is arranged. The end crossover wire **57** extends in the circumferential direction on the other end **20b**. The end crossover wire **57** extends along the circumferential direction over lengths of at least two single coils **51** and one slot.

(89) The inter-pole crossover wire **58** is arranged between the winding finish portion **56** and the end crossover wire **57**. The inter-pole crossover wire **58** is located before the end crossover wire **57** in the winding process, and is therefore also called a front crossover wire. The inter-pole crossover

wire **58** connects between the one end **20a** and the other end **20b** in the axial direction through the inter-pole gap PG which is between the different-phase two single coils **51** separated in the circumferential direction. The inter-pole crossover wire **58** extends between the one end **20a** and the other end **20b** in the axial direction through the inter-pole gap PG which is between two magnetic poles adjacent to each other in the circumferential direction.

(90) The inter-pole crossover wire **59** is arranged between the end crossover wire **57** and the winding start portion **55**. The inter-pole crossover wire **59** is located after the end crossover wire **57** in the winding process, and is therefore also called a rear crossover wire. The inter-pole crossover wire **59** connects between the one end **20a** and the other end **20b** in the axial direction through the inter-pole gap PG which is between the different-phase two single coils **51** separated in the circumferential direction. The inter-pole crossover wire **59** extends between the one end **20a** and the other end **20b** in the axial direction through the inter-pole gap PG which is between two magnetic poles adjacent to each other in the circumferential direction.

(91) In the drawings and the descriptions, the end crossover wire **57**, the winding start portion **55**, the winding finish portion **56**, the inter-pole crossover wire **58**, and the inter-pole crossover wire **59** are assigned identification symbols such as a, b . . . to be identified. The identification symbols such as a, b, . . . correspond to repetition of windings in the winding process. For example, winding of the U-phase winding **50u** starts from a first winding start portion **55a** and ends at a last winding finish portion **56f**. The details of the U-phase winding **50u** will be described below. The V-phase winding **50v** and the W-phase winding **50w** have the similar shape.

(92) With respect to the U-phase winding **50u**, winding of the magnetic pole No. 1 starts from the winding start portion **55a** and ends at a winding finish portion **56a**. In this embodiment, the single coil **51** is wound in a clockwise direction. Alternatively, the single coil **51** may be wound in a counterclockwise direction. In this embodiment, the winding process proceeds rightward from the magnetic pole No. 1. Alternatively, the winding process may proceed leftward from the magnetic pole No. 1.

(93) The single coil **51** has wires that is laminated in several layers in a radial direction of the magnetic pole. The number of layers in the single coil **51** is more than or equal to two, and less than or equal to ten. Preferably, the number of layers in the single coil **51** is more than or equal to two, and less than or equal to seven. In this embodiment, the number of layers in the single coil **51** is more than or equal to two, and less than or equal to five. Specifically, the single coil **51** has three layers. This number of layers defines heights of the base end flange **43** and the top end flange **44**. As a result, the number of layers defines the height TH**40** of the insulator **40**, and consequently defines the height of the rotary electric machine **1**. The number of layers in the single coil **51** is more than or equal to two even in regular winding or random winding. The winding diagram schematically illustrates the single coil **51**, and the number of turns and the number of layers are omitted. The single coil **51** is arranged inside the stator **20**, that is, on an inner circumferential surface of the stator **20**.

(94) The U-phase winding **50u** connects from a winding finish portion **56a** through an inter-pole crossover wire **58a** to an end crossover wire **57a**. The winding start portion **55a** is arranged on the one end **20a**. The end crossover wire **57a** is arranged on the other end **20b**, i.e., an end portion opposite from the conductive member **60**. The multiple conductive members **60** are intensively arranged on the one end **20a** of the stator **20** in the axial direction, and the multiple end crossover wires **57** are intensively arranged on the other end **20b** of the stator **20** in the axial direction. As a result, the both ends of the stator **20** can be effectively used.

(95) The U-phase winding **50u** is continuous from the winding finish portion **56a** to the inter-pole crossover wire **58a**. The inter-pole crossover wire **58a** extends from the one end **20a** to the other end **20b**. The inter-pole crossover wire **58a** extends in the axial direction within the slot S2. The inter-pole crossover wire **58a** extends in the circumferential direction within the slot S2. Thus, the inter-pole crossover wire **58a** extends obliquely within the slot S2. The inter-pole crossover wire

58a is arranged inside the stator **20**. The inter-pole crossover wire **58a** is arranged in the inter-pole gap PG. The inter-pole crossover wire **58a** also functions imperfectly as a part of the single coil **51** in the magnetic pole No. 1.

(96) The U-phase winding **50u** is continuous from the inter-pole crossover wire **58a** to the end crossover wire **57a**. The end crossover wire **57a** extends in the circumferential direction. The end crossover wire **57a** is arranged so as to extend along the circumferential direction of the stator **20**. The end crossover wire **57a** is arranged along the base end flange **43**. The end crossover wire **57a** is arranged outside the stator **20**. The base end flange **43** functions as a stopper that holds the end crossover wire **57a**. A circumferential edge of the base end flange **43** defines a boundary between the inter-pole crossover wire **58a** and the end crossover wire **57a**.

(97) The U-phase winding **50u** is continuous from the end crossover wire **57a** to an inter-pole crossover wire **59a**. The inter-pole crossover wire **59a** extends from the other end **20b** to the one end **20a**. The inter-pole crossover wire **59a** extends in the axial direction within the slot S4. The inter-pole crossover wire **59a** extends in the circumferential direction within the slot S4. Thus, the inter-pole crossover wire **59a** extends obliquely within the slot S4. The inter-pole crossover wire **59a** is arranged inside the stator **20**. The inter-pole crossover wire **59a** also functions imperfectly as a part of the single coil **51** in a magnetic pole No. 4.

(98) The U-phase winding **50u** is continuous from the inter-pole crossover wire **59a** to a winding start portion **55b**. The U-phase winding **50u** repeats a basic structure cycle: the winding start portion **55**; the single coil **51**; the winding finish portion **56**; the inter-pole crossover wire **58**; the end crossover wire **57**; and the inter-pole crossover wire **59**, and finally reaches the last single coil. In the last single coil, the U-phase winding **50u** is continuous from an inter-pole crossover wire **59e** to the winding start portion **55f**. The U-phase winding **50u** is wound on the magnetic pole No. 16 and terminates at the winding finish portion **56f**. The winding finish portion **56f** is connected to the neutral point member **64**.

(99) The U-phase winding **50u** provides the magnetic poles No. 1, 4, 7, 10, 13 and 16. The V-phase winding **50v** and the W-phase winding **50w** have the same shape as the U-phase winding **50u**. Thus, the V-phase winding **50v** provides the magnetic poles No. 2, 5, 8, 11, 14 and 17. The W-phase winding **50w** provides the magnetic poles No. 3, 6, 9, 12, 15 and 18.

(100) The multiple phase windings **50u**, **50v**, **50w** provide two types of multiple inter-pole crossover wires **58**, **59**. These inter-pole crossover wires **58**, **59** are arranged such that (1) only one of them passes obliquely in the slot, (2) both of them cross each other in the slot, or (3) both of them are out of the slot.

(101) (1) In the slot S2, only the inter-pole crossover wire **58** of the U-phase winding **50u** is arranged. In the slot S3, only the inter-pole crossover wire **58** of the V-phase winding **50v** is arranged. In the slot S17, only the inter-pole crossover wire **59** of the V-phase winding **50v** is arranged. In the slot S18, only the inter-pole crossover wire **59** of the W-phase winding **50w** is arranged. In these four slots at the start and end of winding, only one inter-pole crossover wire **58** or only one inter-pole crossover wire **59** is obliquely arranged. These slots S2, S3, S17, S18 are also called both-end slots, in each of which one wire, i.e., the inter-pole crossover wire **58** or the inter-pole crossover wire **59** is arranged.

(102) (2) In the slot S4, the inter-pole crossover wire **58** of the W-phase winding **50w** and the inter-pole crossover wire **59** of the U-phase winding **50u** are arranged. The two inter-pole crossover wires **58**, **59** cross each other in the slot S4, that is, in the inter-pole gap PG. In the slot S5, the inter-pole crossover wire **58** of the U-phase winding **50u** and the inter-pole crossover wire **59** of the V-phase winding **50v** are arranged. The two inter-pole crossover wires **58**, **59** cross each other in the slot S5, that is, in the inter-pole gap PG. In the slot S6, the inter-pole crossover wire **58** of the V-phase winding **50v** and the inter-pole crossover wire **59** of the W-phase winding **50w** are arranged. The two inter-pole crossover wires **58**, **59** cross each other in the slot S6, that is, in the inter-pole gap PG. The crossing arrangements seen in the slots S4, S5, S6 are also seen in the slots

S7, S8, S9, S10, S11, S12, S13, S14, S15, S16 in which two inter-pole crossover wires **58** and **59** of different phase windings are arranged by repeating of the winding process. The two inter-pole crossover wires **58** and **59** cross each other at the center of the inter-pole gap PG in the axial direction. The two inter-pole crossover wires **58** and **59** cross each other at the center of the inter-pole gap PG in the circumferential direction. These slots S4 to S16 are also called intermediate slots, in each of which two wires, i.e., the inter-pole crossover wire **58** and the inter-pole crossover wire **59** are arranged to cross each other.

(103) (3) In slot S1, neither the inter-pole crossover wire **58** nor the inter-pole crossover wire **59** is disposed in the slot S1. The slot S1 is also referred to as a boundary slot, in which the inter-pole crossover wire is not arranged.

(104) The end crossover wire **57** is arranged on the other end **20b**. The end crossover wire **57** is arranged along the base end flange **43**. The end crossover wire **57** is located radially outward of the base end flange **43**. The end crossover wires **57** are arranged so as to overlap each other in the radial direction and the axial direction. In this embodiment, the end crossover wires **57** of different phases are stacked in the axial direction on the other end **20b**. The number of the end crossover wires **57** arranged on the other end **20b** is two or less at every positions in the circumferential direction. In the drawings, the symbols *1 and *2 indicate the number of the end crossover wires **57** at the positions. On the axial end portion (the other end **20b**) of the stator **20**, the multiple end crossover wires **57** are arranged so that (1) only one of them extends, (2) two of them extend to overlap each other, or (3) none of them extends.

(105) (1) The number of the end crossover wires **57** located along the base end flange **43** of the magnetic pole No. 2 is one. The number of the end crossover wires **57** located along the base end flange **43** of the magnetic pole No. 17 is one. The number of the end crossover wires **57** overlapping the slot S3 in the axial direction is one. This is because the inter-pole crossover wire **58** of the V-phase winding **50v** is arranged in the slot S3. The number of the end crossover wires **57** overlapping the slot S4 in the axial direction is one. This is because two of the inter-pole crossover wires **58**, **59** are arranged in the slot S4. The number of the end crossover wires **57** overlapping the slot S5 in the axial direction is one. This is because two of the inter-pole crossover wires **58**, **59** are arranged in the slot S5. The presence of the one end crossover wire **57** at the ends of the slots S3, S4, S5 is repeatedly provided also to the slots S6, S7, S8, S9, S10, S11, S12, S13, S14, S15, S16, S17.

(106) (2) The number of the end crossover wires **57** located along the base end flange **43** of the magnetic pole No. 3 is two. The number of the end crossover wires **57** located along the base end flange **43** of the magnetic pole No. 4 is two. The number of the end crossover wires **57** located along the base end flange **43** of the magnetic pole No. 5 is two. The presence of the multiple end crossover wires **57** on the base end flanges **43** of the magnetic poles No. 3, 4, 5 is repeatedly provided also to the base end flanges **43** of the magnetic poles No. 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16.

(107) (3) There is no end crossover wire **57** overlapping the slot S1 in the axial direction (zero). There is no end crossover wire **57** located along the base end flange **43** of the magnetic pole No. 1. There is no end crossover wire **57** located along the base end flange **43** of the magnetic pole No. 18. This is because the boundary slot S1 and the other ends **20b** of the both end magnetic poles do not require a crossover wire. There is no end crossover wire **57** overlapping the slot S2 in the axial direction. This is because the inter-pole crossover wire **58** functioning as a crossover wire is arranged in the slot S2. There is no end crossover wire **57** overlapping the slot S18 in the axial direction. This is because the inter-pole crossover wire **59** functioning as a crossover wire is arranged in the slot S18.

(108) In this embodiment, the crossover wire **54** includes one inter-pole crossover wire **58**, one inter-pole crossover wire **59**, and one end crossover wire **57**. One inter-pole crossover wire **58** extends from the winding finish portion **56** of one of the two same-phase single coils **51**. One inter-

pole crossover wire **59** extends from the winding start portion **55** of another of the two same-phase single coils **51**. The crossover wire **54** includes one inter-pole crossover wire **58**, the end crossover wire **57**, and one inter-pole crossover wire **59**, in this order. Each of the multiple phase windings **50u**, **50v**, **50w** forming the multi-phase winding has inter-pole crossover wires **58**, **59**. As a result, in some of the inter-pole gaps PG, the inter-pole crossover wires **58**, **59** of different phases cross each other. The multiple slots S1 to S18 formed between the multiple magnetic poles include one slot S1 in which there is no inter-pole crossover wire **58**, **59**. The multiple slots S1 to S18 include multiple slots S2, S3, S17, S18 in which one inter-pole crossover wire **58** or one inter-pole crossover wire **59** is arranged. The multiple slots S1 to S18 include multiple slots S4, S5, S6, S7, S8, S9, S10, S11, S12, S13, S14, S15, S16 in which two inter-pole crossover wires **58**, **59** are arranged so as to cross each other. The end crossover wire **57** is arranged on one of inner and outer surfaces of the stator **20** in the radial direction, which is opposite from the surface on which the single coil **51** is arranged. In other words, the magnetic pole includes the base end flange **43**, and the end crossover wire **57** and the inter-pole crossover wires **58**, **59** are arranged on different sides of the base end flange **43** in the radial direction of the rotary electric machine **1**.

(109) A method for manufacturing the rotary electric machine includes a step of assembling the rotor **10** and a step of assembling the stator **20**. The step of assembling the stator **20** includes a step of assembling the stator core **30**, a step of mounting the insulator **40** on the stator core **30**, a step of winding the coil **50**, and a step of forming the multiple connection portions **65**, **66**, **67**, **68**, **69**. In the step of winding the coil **50**, a winding machine is used to wind a wire around the stator core **30** with the insulator **40** so as to form the coil **50**. This step can be carried out by one winding nozzle in sequence or by multiple winding nozzles in parallel.

(110) In the step of winding the coil **50**, the winding starts from the one end **20a** and the first single coil **51** is wound. This step proceeds while the end crossover wire **57** is arranged on the other end **20b**. In a repeating step, the single coil **51** is started to be wound from the other end **20b**, and the single coils **51** are successively wound. Moreover, at this step, the single coils **51** are wound one after another while both the inter-pole crossover wire **58** and the inter-pole crossover wire **59** are arranged in the slots. In this step, winding of the last single coil **51** is completed on the one end **20a**.

(111) In the step of forming the multiple connection portions **65**, **66**, **67**, **68**, **69**, the multiple coil ends **52** are connected to the multiple conductive members **60**. In this step, the conductive member **60** and the coil end **52** are electrically connected. This step can be performed in a manner: the conductive member **60** is arranged in the inter-pole gap PG; the coil end **52** is arranged so as to be in contact with the conductive member **60**; the joint portion **60d** is bent so as to wrap around the coil end **52**; and fusing is performed. Alternatively, this step may be performed in a manner: the coil end **52** is arranged so as to be in contact with the conductive member **60** outside the inter-pole gap PG; the joint portion **60d** is bent so as to wrap around the coil end **52**; fusing is performed; and the conductive member **60** is arranged in the inter-pole gap PG.

(112) Further, the method for manufacturing the rotary electric machine includes a step of fixing the multiple conductive members **60**. This step can be performed before or after the step of forming the multiple connection portions. In this embodiment, the terminal block **80** positions and fixes the multiple power end members **61**, **62**, **63** at predetermined positions on the stator **20**.

(113) According to the embodiment described above, at least one inter-pole crossover wire **58** or inter-pole crossover wire **59** is provided. Thus, the number of the end crossover wires **57** on the axial end portion of the slot (inter-pole gap PG) in which the inter-pole crossover wire **58** or the inter-pole crossover wire **59** is arranged can be reduced. This is because the inter-pole crossover wires **58**, **59** provides the function of the end crossover wire **57** arranged on the axial end portion of the slot (inter-pole gap PG). In particular, the inter-pole crossover wire **58** and the inter-pole crossover wire **59** of different phase windings cross each other in the slot (inter-pole gap PG). Thus, the number of the end crossover wires **57** on the axial end portion of the slot (inter-pole gap PG) in

which the inter-pole crossover wire **58** and the inter-pole crossover wire **59** are arranged can be reduced to the number of phases—2. In the case of a three-phase winding, the number of end crossover wires **57** on the axial end portion of the slot (inter-pole gap PG) in which the inter-pole crossover wire **58** and the inter-pole crossover wire **59** are arranged can be reduced to one. As a result, a compact rotary electric machine **1** is provided.

(114) According to the embodiment described above, the connection portions **65, 66, 67, 68, 69** for the multiple coil ends **52** can be arranged in the inter-pole gaps PG. Therefore, miniaturization of the connection portions **65, 66, 67, 68, 69** for the coil ends **52** in the axial direction is achieved. According to this embodiment, the connection portions **65, 66, 67, 68, 69** can be easily manufactured. According to this embodiment, a rotary electric machine having a short length in the axial direction is provided.

Second Embodiment

(115) This embodiment is a modification of the basic preceding embodiment. In the above embodiment, the rotary electric machine **1** includes the neutral point member **64** and the arc-shaped terminal block **80**. Instead, in this embodiment, a rotary electric machine **1** includes a multi-terminal neutral point member **264** and an annular terminal block **280**. In this embodiment, the elements corresponding to those in the above-described embodiment are assigned with the same reference numerals. The descriptions of the above embodiment can be referred to for the descriptions of the elements assigned with the same reference numerals. In this embodiment, the rotary electric machine **1** includes a rotor **10** having 20 poles and a stator **20** having 15 poles.

(116) In FIGS. 5 to 12, particularly in FIG. 9, the neutral point member **264** has three connection portions **68, 69, 270**. The neutral point member **264** provides three terminals for a three phase winding. The connection portion **68** provides electrical connection between a coil end of a first phase winding of the three phase winding and the neutral point member **264**. The connection portion **69** provides electrical connection between a coil end of a second phase winding of the three phase winding and the neutral point member **264**. The connection portion **270** provides electrical connection between a coil end of a third phase winding of the three phase winding and the neutral point member **264**.

(117) The multiple connection portions **65, 66, 67, 68, 69, 270** are dispersively arranged in the multiple inter-pole gaps PG next to each other. The multiple connection portions **65, 66, 67, 68, 69, 270** are arranged in the multiple inter-pole gaps PG next to each other, in a one-to-one relationship therebetween. In this embodiment, one connection portion is arranged in one inter-pole gap PG. As a result, the six connection portions **65, 66, 67, 68, 69, 270** are arranged in the six inter-pole gaps PG next to each other. The neutral point member **264** has a shape that can be called a comb teeth type or a pitchfork type. The neutral point member **264** includes an arc portion that extends in the circumferential direction along a yoke **32** of a stator core **30**, and radial portions that extends in the radial direction inward from the arc portion as a connection portion. The radial portions are disposed at equal intervals in the circumferential direction.

(118) Also in this embodiment, the multiple conductive members **60** include a circumferential extending portion **60a** and a radial extending portion **60b**. For example, the power end members **61, 63** have the circumferential extending portion **60a** and the radial extending portion **60b**. The power end member **62** includes only the radial extending portion **60b**. The neutral point member **264** includes a circumferential extending portion **60a** and three radial extending portions **60b**.

(119) In the above-described embodiment, the terminal block **80** supports only the multiple power end members **61, 62, 63**. The terminal block **80** does not support the neutral point member **64**. In contrast, the terminal block **280** supports all of the multiple conductive members **60**. The terminal block **280** supports both the multiple power end members **61, 62, 63** and the neutral point member **264**. The multiple conductive members **60** are insert-molded on the terminal block **280**.

(120) In FIGS. 5 to 12, in particular FIGS. 5, 6, 7 and 8, the terminal block **280** has an annular shape. The terminal block **280** extends in the circumferential direction along the yoke **32**. The

terminal block **280** is located outward of the base end flange **43** in the radial direction. Also in this embodiment, the terminal block **280** provides a body portion **281** and a connector portion **82**. The body portion **281** has an annular shape. The connector portion **82** is positioned at a part of the annular body portion **281**.

(121) As shown in FIGS. **10** and **11**, an insulator **40** has a protrusion **245**. The insulator **40** has multiple protrusions **245**. The protrusion **245** further projects outward in the radial direction from a radially outer surface of an end portion of the base end flange **43** in the axial direction. The protrusion **245** functions as a stopper for positioning and holding multiple crossover wires **54** along the base end flange **43**. Arrangement and the number of the multiple crossover wires **54** are shown in winding diagrams described later.

(122) In FIG. **13**, the neutral point member **264** connects the multiple phase windings **50u**, **50v**, **50w**. The coil **50** is the same as in the above embodiment.

(123) Also in this embodiment, the number of end crossover wires **57** on the axial end portion of the slot (inter-pole gap PG) in which the inter-pole crossover wire **58** and the inter-pole crossover wire **59** are arranged can be reduced. As a result, a compact rotary electric machine **1** is provided. Also in this embodiment, the connection portions **65**, **66**, **67**, **68**, **69**, **270** for the multiple coil ends **52** can be arranged in the inter-pole gaps PG. Therefore, miniaturization of the connection portions **65**, **66**, **67**, **68**, **69**, **270** for the multiple coil ends **52** in the axial direction is achieved. According to this embodiment, the connection portions **65**, **66**, **67**, **68**, **69**, **270** can be easily manufactured. According to this embodiment, a rotary electric machine having a short length in the axial direction is provided.

Third Embodiment

(124) This embodiment is a modification of the basic preceding embodiments. In the above-described embodiments, the rotary electric machine **1** includes both the inter-pole crossover wires **58** and the inter-pole crossover wires **59**. Alternatively, in this embodiment, a rotary electric machine **1** includes only inter-pole crossover wires **58**.

(125) In FIG. **14**, the rotary electric machine **1** includes a coil **350**. The coil **350** has only the inter-pole crossover wires **58** in multiple slots. As a result, the number of end crossover wires **57** is reduced on the end portion of a slot in the axial direction in which the inter-pole crossover wire **58** is arranged. The number of end crossover wires **57** is reduced to two or less.

(126) In this embodiment, a crossover wire **54** includes one inter-pole crossover wire **58** extending from the winding finish portion **56** of one of the two same-phase single coils **51**, and the end crossover wire **57** only. Also in this embodiment, a compact rotary electric machine **1** is provided.

Fourth Embodiment

(127) This embodiment is a modification of the basic preceding embodiments. In the above-described embodiments, the rotary electric machine **1** includes the inter-pole crossover wires **58** at least. Alternatively, in this embodiment, a rotary electric machine **1** includes only inter-pole crossover wires **59**.

(128) In FIG. **15**, the rotary electric machine **1** includes a coil **450**. The coil **450** has only the inter-pole crossover wires **59** in multiple slots. As a result, the number of end crossover wires **57** is reduced on the end portion of a slot in the axial direction in which the inter-pole crossover wire **59** is arranged. The number of end crossover wires **57** is reduced to two or less.

(129) In this embodiment, a crossover wire **54** includes one inter-pole crossover wire **59** extending from the winding start portion **55** of another of the two same-phase single coils **51**, and the end crossover wire **57** only. Also in this embodiment, a compact rotary electric machine **1** is provided.

Fifth Embodiment

(130) This embodiment is a modification of the basic preceding embodiments. In the above-described embodiments, the rotary electric machine **1** has the end crossover wires **57** arranged outward of the base end flange **43** in the radial direction. Additionally, in this embodiment, a base end flange **43** provides multiple communication portions **548**, **549** for end crossover wires **57**. In

this embodiment, a stator **20** provides fifteen magnetic poles.

(131) In FIG. **16**, the stator **20** includes a stator core **30**, an insulator **40**, and a coil **50**. In the figure, the end crossover wires **57**, inter-pole crossover wires **58**, and inter-pole crossover wires **59** in the stator **20** are illustrated. In order to facilitate understanding and to avoid complicating the drawing, the coil **50** is schematically illustrated and is partially illustrated by a hidden line (dash line). The insulator **40** includes the base end flange **43** and a top end flange **44**. The base end flange **43** and the top end flange **44** define a range of the single coil **51**. Further, the base end flange **43** includes multiple communication portions for defining positions of wires of the coil **50**. The multiple communication portions provide slit openings in the radial direction. The multiple communication portions allow the wires to be arranged. The multiple communication portions are located at both ends of the end crossover wire **57** and define a range of the end crossover wire **57** in the circumferential direction. The coil **50** includes multiple phase windings **50u**, **50v**, **50w**. The coil **50** is wound from the three magnetic poles illustrated in the center of the figure.

(132) The base end flange **43** is provided for each magnetic pole on one end **20a**. The base end flange **43** includes a first flange piece **546** and a second flange piece **547** on the other end **20b** for providing the multiple communication portions. The first flange piece **546** is located outward of each magnetic pole in the radial direction and protrudes in the axial direction. The first flange piece **546** is also called a magnetic pole flange. The second flange piece **547** is located outward of each slot S1 to S15 in the radial direction and protrudes in the axial direction. The second flange piece **547** has a protrusion **245**. The second flange piece **547** is also called a slot flange. The first flange pieces **546** and the second flange pieces **547** are alternately provided along the circumferential direction. The first flange pieces **546** and the second flange pieces **547** define therebetween first communication portions **548** and second communication portions **549** alternately. With respect to the processing direction of the winding process, the first communication portion **548** is located frontward of the first flange piece **546**, and the second communication portion **549** is located rearward of the first flange piece **546**. In other words, the second communication portion **549** is located frontward of the second flange piece **547**, and the first communication portion **548** is located rearward of the second flange piece **547**.

(133) The first communication portion **548** and the second communication portion **549** are located at different ends of the end crossover wire **57**, respectively. The first communication portion **548** is located between the inter-pole crossover wire **58** and the end crossover wire **57** and defines the boundary between them. The first communication portion **548** is located at the finish end of the inter-pole crossover wire **58**. The first communication portion **548** is located at the start end of the end crossover wire **57**. The first communication portion **548** is also an outlet communication portion that draws the wire from inside to outside the stator **20**. The second communication portion **549** is located between the end crossover wire **57** and the inter-pole crossover wire **59** and defines the boundary between them. The second communication portion **549** is located at the start end of the inter-pole crossover wire **59**. The second communication portion **549** is located at the finish end of the end crossover wire **57**. The second communication portion **549** is also an inlet communication portion that draws the wire from outside to inside the stator **20**. In this embodiment, two inter-pole crossover wires **58**, **59** are arranged in the slot. The second communication portion **549** located frontward of the second flange piece **547** that is the slot flange provides an inlet for drawing the wire from outside to inside of the stator **20**. The first communication portion **548** located rearward of the second flange piece **547** that is the slot flange provides an outlet for drawing the wire from inside to outside of the stator **20**. When the rotary electric machine **1** includes both the inter-pole crossover wire **58** and the inter-pole crossover wire **59**, both the first communication portion **548** and the second communication portion **549** are used. When the rotary electric machine **1** includes the inter-pole crossover wire **58** or the inter-pole crossover wire **59**, only the first communication portion **548** or only the second communication portion **549** are used. For example, when the rotary electric machine **1** includes only the inter-pole

crossover wire **58**, only the first communication portion **548** is used as the inlet and the outlet. For example, when the rotary electric machine **1** includes only the inter-pole crossover wire **59**, only the second communication portion **549** is used as the inlet and the outlet.

(134) The U-phase winding **50u** will be described as a representative example. The U-phase winding **50u** starts to be wound from the winding start portion **55a** on the first magnetic pole and is continuous from the winding finish portion **56a** to the inter-pole crossover wire **58a**. The inter-pole crossover wire **58a** is drawn outward of the base end flange **43** in the radial direction through the first communication portion **548**, and is continuous to the end crossover wire **57a**. The end crossover wire **57a** is drawn inward of the base end flange **43** in the radial direction through the second communication portion **549**, and is continuous to the inter-pole crossover wire **59a**. The multiple phase windings **50u**, **50v**, **50w** have similar shapes. Therefore, in some slots, the inter-pole crossover wire **58** of one phase winding intersects with the inter-pole crossover wire **59** of another phase winding. For example, in the slot S4, the inter-pole crossover wire **58** of the W-phase winding **50w** and the inter-pole crossover wire **59** of the U-phase winding **50u** cross each other. In this embodiment, since fifteen magnetic poles are provided, the coil **50** includes elements identified by the a, b, c, d, e identifiers. For example, between the magnetic pole No. 7 and the magnetic pole No. 10, an inter-pole crossover wire **58c**, an end crossover wire **57c**, and an inter-pole crossover wire **59c** are arranged. The end crossover wire **57c** extends from the first communication portion **548** as an outlet to the second communication portion **549** as an inlet.

(135) In FIG. 17, one phase winding has the single coils **51** positioned on both sides of the inter-pole crossover wire **58**, the end crossover wire **57**, and the inter-pole crossover wire **59**. The coil **50** is illustrated as viewed from an inside of the stator **20**. Therefore, in one slot, two wires of different phase windings, i.e., the inter-pole crossover wire **58** and the inter-pole crossover wire **59** cross each other. As a result, the number of the end crossover wires **57** outside the base end flange **43** is reduced, and the compact rotary electric machine **1** is provided.

Sixth Embodiment

(136) This embodiment is a modification of the basic preceding embodiments. In the above-described embodiments, the rotary electric machine **1** includes both the inter-pole crossover wires **58** and the inter-pole crossover wires **59**. Additionally, in this embodiment, a rotary electric machine **1** includes only inter-pole crossover wires **58**.

(137) In FIG. 18, a stator **20** includes an insulator **40** and a coil **350** described in the fifth embodiment. The coil **350** includes only inter-pole crossover wires **58**. In this embodiment, only first communication portions **548** are used as inlets and outlets. For example, the U-phase winding **50u** is drawn outward from one first communication portion **548** and is drawn inward from another first communication portion **548**. In this configuration, since the coil **350** is formed only by the inter-pole crossover wires **58**, the end crossover wires **57** become slightly long. For example, an end crossover wire **57c** extends to a first communication portion **548** immediately before the magnetic pole No. 10.

(138) In FIG. 19, the coil **350** includes only inter-pole crossover wires **58**. Instead of the coil **350**, a coil **450** including only the inter-pole crossover wires **59** may be provided. Also in this embodiment, a compact rotary electric machine **1** is provided.

Seventh Embodiment

(139) This embodiment is a modification of the basic preceding embodiments. In the preceding embodiments, the inter-pole crossover wires **58**, the inter-pole crossover wires **59** or both the wires **58** and **59** are obliquely arranged in the slots. Alternatively, in the present embodiment, inter-pole crossover wires **58** or inter-pole crossover wires **59** can be provided in a variety of arrangements.

(140) In the following descriptions, the rotary electric machine **1** is described with use of an axial direction AD, a radial direction RD, and a circumferential direction CD. An element belonging to one phase of a multi-phase winding may be assigned a numeral with an identifier g, h or i. An element belonging to another phase of the multi-phase winding may be assigned a numeral with an

identifier j, k, L or m. An element belonging to further another phase of the multi-phase winding may be assigned a numeral with an identifier p, q, r or s. A stator **20** includes single coils **51g**, **51q**, **51k**, **51h**, **51r**, **51L**, **51i**, **51s** and **51m** in this order. Similarly, an inter-pole crossover wire **58g** or an inter-pole crossover wire **58k** will be described as a representative of the inter-pole crossover wires **58**. Similarly, an inter-pole crossover wire **59p** or an inter-pole crossover wire **59g** will be described as a representative of the inter-pole crossover wires **59**.

(141) Also in the present embodiment, each of crossover wires **54** includes an end crossover wire **57**, an inter-pole crossover wire **58**, and a inter-pole crossover wire **59**. In the present embodiment, each inter-pole crossover wire **58**, **59** includes a transition portion (e.g., second portion **758g2**, **759p2**, first portion **758g1**, **759p1**) between the inter-pole crossover wire **58**, **59** and the end crossover wire **57**.

(142) In FIGS. **20** and **22**, the stator **20** includes multiple pseudo teeth **725**. The pseudo teeth **725** are positioned between two adjacent magnetic poles in the circumferential direction. Pseudo teeth **725** are positioning members for fixing positions of the inter-pole crossover wires **58** and positions of inter-pole crossover wires **59**. The stator **20** includes three pseudo teeth **725**. Each pseudo tooth **725** is positioned between two adjacent teeth **31**. The three pseudo teeth **725** are separated from each other by 72° in mechanical angle. The three pseudo teeth **725** are separated from each other by 720° in electrical angle (integer multiple of 360°). The pseudo teeth **725** is capable of adjusting a torque fluctuation observed in a rotor **10** to have a desired waveform. The stator **20** has first slots Sa in which the pseudo teeth **725** are provided and second slots Sb in which the pseudo teeth **725** are not provided. Each pseudo tooth **725** has an end surface facing in the axial direction, and a width of the end surface in the circumferential direction is $\frac{1}{3}$ or less of a width of a first slot Sa in the circumferential direction. An inter-pole crossover wire **58** and an inter-pole crossover wire **59** bypass the pseudo tooth **725** in the first slot Sa.

(143) FIG. **21** is an enlarged view of a part indicated by an arrow XXI in FIG. **20**. FIG. **23** is an enlarged view of a part indicated by an arrow XXIII in FIG. **22**. Each pseudo tooth **725** includes a magnetic material. The pseudo tooth **725** is provided primarily by a magnetic core that is a material continuous from the stator core **30**. The pseudo tooth **725** includes an insulator **740a**. The insulator **740a** is a material continuous from an insulator **40**. Therefore, the insulator **740a** provides electrical insulation in the pseudo tooth **725**. The inter-pole crossover wires **58**, **59** are both arranged so as to contact the pseudo teeth **725**. The pseudo tooth **725** is a positioning member for fixing the positions of the inter-pole crossover wires **58**, **59**.

(144) In the first slot Sa, the inter-pole crossover wire **58g** is provided as an inter-pole crossover wire **758** positioned in the first slot Sa. A single coil **51g** has a winding finish portion **56g**. The winding finish portion **56g** is arranged at a tip end portion of the single coil **51g**, that is, an inner portion in the radial direction. The inter-pole crossover wire **58g** extends diagonally from the winding finish portion **56g** of the single coil **51g** in a slot space between the single coil **51g** and the pseudo tooth **725**. The inter-pole crossover wire **58g** extends from the winding finish portion **56g** in the circumferential direction, the radial direction, and the axial direction. The inter-pole crossover wire **58g** extends through the air. The inter-pole crossover wire **58g** is arranged so as to contact the end surface of the pseudo tooth **725** that faces in the axial direction. Further, the inter-pole crossover wire **58g** extends on an end surface of the stator core **30** that faces in the axial direction, and then reaches an end portion of a base end flange **43q** that faces in the radial direction. In other words, the inter-pole crossover wire **58g** reaches an outer side of the base end flange **43q** in the radial direction. As a result, the inter-pole crossover wire **58g** is connected to the end crossover wire **57g**.

(145) The inter-pole crossover wire **58g** includes the first portion **758g1** and the second portion **758g2**. The first portion **758g1** is obliquely arranged in a partial slot Sa1 that is between the single coil **51g** and the pseudo tooth **725**. The second portion **758g2** is arranged outward of a partial slot Sa2 in the axial direction, the partial slot Sa2 is between the pseudo tooth **725** and a single coil **51q**.

The first portion **758g1** is greater than the second portion **758g2** in lengths in the axial direction and the radial direction. Since the second portion **758g2** is located on an end surface of the stator **20** that faces in the axial direction, the second portion **758g2** may be classified as a part of the end crossover wire **57g**. In this case, the inter-pole crossover wire **58g** is provided only as the first portion **758g1**. The second portion **758g2** is also referred to as a transition portion positioned between the end crossover wire **57g** and the first portion **758g1** as the inter-pole crossover wire **58g**.

(146) In the first slot **Sa**, the inter-pole crossover wire **59p** is provided as an inter-pole crossover wire **759** positioned in the first slot **Sa**. An end crossover wire **57p** is arranged on an end portion of a base end flange **43g** facing in the axial direction, that is, on an outer side of the base end flange **43g** in the axial direction. The inter-pole crossover wire **59p** extends from the end crossover wire **57p** and contacts the end surface of the pseudo tooth **725** that faces in the axial direction. Further, the inter-pole crossover wire **59p** extends from the end surface of the pseudo tooth **725**, and the inter-pole crossover wire **59p** obliquely crosses a space in the first slot **Sa**. The single coil **51q** has a winding start portion **55q**. The winding start portion **55q** is arranged at a base end portion of the single coil **51q**, that is, an outer portion of the single coil **51q** in the radial direction. The inter-pole crossover wire **59p** extends from the end surface of the pseudo tooth **725** toward winding start portion **55q** in the circumferential direction and the axial direction. The inter-pole crossover wire **59p** extends through the air. The inter-pole crossover wire **59p** extends from the end surface of the pseudo tooth **725** and reaches the winding start portion **55q**. As a result, the inter-pole crossover wire **59p** is connected to the winding start portion **55q** of the single coil **51q**.

(147) The inter-pole crossover wire **58g** and the inter-pole crossover wire **59p** cross with each other on the pseudo tooth **725**. The two single coils **51g** and **51q** are adjacent to each other in the circumferential direction. The inter-pole crossover wire **58g** extending from one single coil **51g** and the inter-pole crossover wire **59p** extending from the other single coil **51q** cross each other in an inter-pole gap **PG**. The inter-pole crossover wire **59p** includes the first portion **759p1** and the second portion **759p2**. The first portion **759p1** is arranged outward of the partial slot **Sa1** in the axial direction, the partial slot **Sa1** is between the pseudo tooth **725** and the single coil **51g**. The second portion **759p2** is obliquely arranged in the partial slot **Sa2** that is between the single coil **51q** and the pseudo tooth **725**. The second portion **759p2** is greater than the first portion **759p1** in length in the axial direction. Since the first portion **759p1** is located on the end surface of the stator **20** that faces in the axial direction, the first portion **759p1** may be classified as a part of the end crossover wire **57p**. In this case, the inter-pole crossover wire **59p** is provided only as the second portion **759p2**. The first portion **759p1** is also referred to as a transition portion positioned between the end crossover wire **57p** and the second portion **759p2** as the inter-pole crossover wire **59p**.

(148) FIG. **24** shows a cross section taken along a line **XXIV-XXIV** in FIG. **20**. FIG. **25** is an enlarged view of a part indicated by an arrow **XXV** in FIG. **24**. In FIG. **25**, insulators **740a** and **740b** are arranged on both end surfaces of the pseudo tooth **725** in the axial direction. Arranged on both end surfaces of the pseudo teeth **725** in the axial direction are an end crossover wire **57j** that is for one phase, the inter-pole crossover wire **58g** that is for another phase, and the inter-pole crossover wire **59p** that is for another phase. The inter-pole crossover wire **58g** and the inter-pole crossover wire **59p** are shifted from the end crossover wire **57j** in the radial direction. The end crossover wire **57j**, the inter-pole crossover wire **58g** and the inter-pole crossover wire **59p** are arranged in the radial direction on the pseudo tooth **725**. The end crossover wire **57j**, the inter-pole crossover wire **58g** and the inter-pole crossover wire **59p** are arranged within a range of a height of a coil end of the single coil **51g** in the axial direction of the pseudo tooth **725**.

(149) FIG. **26** is an enlarged view of a part indicated by an arrow **XXVI** in FIG. **24**. In FIG. **26**, a cross section of a single coil **51m** is illustrated. An end crossover wire **57i** and an end crossover wire **57s** are arranged outward of a base end flange **43m** of a single coil **51m** in the radial direction. The end crossover wire **57i** connects a single coil **51i** and a single coil of the same phase (not

shown). The end crossover wire **57s** connects a single coil **51s** and a single coil of the same phase (not shown). The base end flange **43m** is a retaining member for retaining the two end crossover wires **57i**, **57s**. The end crossover wire **57i** and the end crossover wire **57s** are arranged outward of a magnetic pole in the radial direction, that is, outward of the base end flange **43m** in the radial direction. The end crossover wire **57i** and the end crossover wire **57s** are arranged within a range of a height of a coil end of the single coil **51g** in the axial direction.

(150) The first portion **758g1** of the inter-pole crossover wire **58g** extends through the inter-pole gap PG and is located between one end **20a** and the other end **20b** of the stator **20** in the axial direction. The first portion **758g1** extends at least in the circumferential direction and obliquely in the axial direction. As a result, the number of crossover wires **54** stacked in the axial direction or the radial direction on an outer side of the insulator **40** in the radial direction is reduced. The second portion **759p2** of the inter-pole crossover wire **59p** extends through the inter-pole gap PG and is located between the one end **20a** and the other end **20b** of the stator **20** in the axial direction. The second portion **759p2** extends at least in the circumferential direction and obliquely in the axial direction. As a result, the number of crossover wires **54** stacked in the axial direction or the radial direction on an outer side of the insulator **40** in the radial direction is reduced. The inter-pole crossover wire **58g** and the inter-pole crossover wire **59p** cross each other in the inter-pole gap PG on an inner side of the base end flanges **43g** and **43q** in the radial direction. Therefore, the crossover wires **54** are shifted from each other in the radial direction within an angle range of the first slot Sa. As a result, the number of crossover wires **54** stacked in the axial direction or the radial direction on an outer side of the insulator **40** in the radial direction is reduced. The number of crossover wires **54** stacked in the axial direction or the radial direction on an outer side of the insulator **40** in the radial direction is limited to two.

(151) FIG. **27** is an enlarged view of a part indicated by an arrow XXVII in FIG. **20**. FIG. **28** is an enlarged view of a part indicated by an arrow XXVIII in FIG. **22**. In the second slot Sb, the inter-pole crossover wire **58k** is provided as an inter-pole crossover wire **758** positioned in the second slot Sb. The inter-pole crossover wire **58k**, a first portion **758k1** and a second portion **758k2** shown in FIGS. **27** and **28** have the same shapes as the inter-pole crossover wire **58g**, the first portion **758g1** and the second portion **758g2** shown in FIGS. **21** and **23**, respectively. In the second slot Sb, the inter-pole crossover wire **59g** is provided as an inter-pole crossover wire **759** positioned in the second slot Sb. The inter-pole crossover wire **59g**, the first portion **759g1** and the second portion **759g2** shown in FIGS. **27** and **28** have the same shapes as the inter-pole crossover wire **59p**, the first portion **759p1** and the second portion **759p2** shown in FIGS. **21** and **23**, respectively. The inter-pole crossover wire **58k** and the inter-pole crossover wire **59g** cross each other in an end space (i.e., inter-pole gap PG) of the second slot Sb in the axial direction. The inter-pole crossover wire **58k** and the inter-pole crossover wire **59g** are curved and arranged as if the pseudo tooth **725** was virtually present there.

(152) FIG. **29** is an enlarged view of a part indicated by an arrow XXIX in FIG. **27**. FIG. **29** shows a cross section in the second slot Sb in which a pseudo tooth **725** is not provided. Arranged in the second slot Sb, are an end crossover wire **57q** that is for one phase, the inter-pole crossover wire **58k** that is for another phase, and the inter-pole crossover wire **59g** that is for another phase. The inter-pole crossover wire **58k** and the inter-pole crossover wire **59g** are shifted from the end crossover wire **57q** in the radial direction. The end crossover wire **57q**, the inter-pole crossover wire **58k** and the inter-pole crossover wire **59g** are arranged in the radial direction. The end crossover wire **57q**, the inter-pole crossover wire **58k** and the inter-pole crossover wire **59g** are arranged within a range of a height of a coil end of a single coil **51k** in a middle portion of the second slot Sb in the circumferential direction.

(153) As shown in FIG. **28**, the first portion **758k1** of the inter-pole crossover wire **58k** extends through the inter-pole gap PG and is located between the one end **20a** and the other end **20b** of the stator **20** in the axial direction. The first portion **758k1** extends at least in the circumferential

direction and obliquely in the axial direction. As a result, the number of crossover wires **54** stacked in the axial direction or the radial direction on an outer side of the insulator **40** in the radial direction is reduced. The second portion **759g2** of the inter-pole crossover wire **59g** extends through the inter-pole gap PG and is located between one end **20a** and the other end **20b** of the stator **20** in the axial direction. The second portion **759g2** extends at least in the circumferential direction and obliquely in the axial direction. As a result, the number of crossover wires **54** stacked in the axial direction or the radial direction on an outer side of the insulator **40** in the radial direction is reduced. The inter-pole crossover wire **58k** and the inter-pole crossover wire **59g** cross each other in the inter-pole gap PG on an inner side of the base end flanges **43k** and **43h** in the radial direction. Therefore, the crossover wires **54** are shifted from each other in the radial direction within an angle range of the first slot Sa. As a result, the number of crossover wires **54** stacked in the axial direction or the radial direction on an outer side of the insulator **40** in the radial direction is reduced. The number of crossover wires **54** stacked in the axial direction or the radial direction on an outer side of the insulator **40** in the radial direction is limited to two.

(154) Also in the present embodiment, since the inter-pole crossover wires are provided, a volume for the crossover wires can be reduced. In the illustrated example, the number of stacked crossover wires can be reduced in the angle ranges of the slots Sa and Sb in the circumferential direction.

When the crossover wires are stacked in the radial direction of the stator **20**, the size of the stator **20** in the radial direction can be reduced. When the crossover wires are stacked in the axial direction of the stator **20**, the size of the stator **20** in the axial direction can be reduced.

Furthermore, the crossover wires is positioned within a range of the coil ends of the single coils, that is, within the range of the heights of the single coils in the axial direction. Thereby, the height of the stator **20** in the axial direction can be reduced.

Eighth Embodiment

(155) This embodiment is a modification of the basic preceding embodiments. In the previous embodiments, the inter-pole crossover wires **58**, **59** are arranged on one side of the stator **20** in the axial direction. In other words, the inter-pole crossover wires **58**, **59** are arranged on one side of a pseudo tooth **725** in the axial direction. Instead, in the present embodiment, inter-pole crossover wires **58**, **59** are separately arranged on both sides of a stator **20** in the axial direction.

(156) FIGS. **30**, **31**, and **32** correspond to FIGS. **21**, **23**, and **25**, respectively. In these drawings, an inter-pole crossover wire **58g** and an inter-pole crossover wire **59p** are arranged on opposite sides of the stator **20** in the axial direction. The inter-pole crossover wire **58g** and the inter-pole crossover wire **59p** are arranged at opposite ends of a pseudo tooth **725** in the axial direction. In the first slot Sa, the inter-pole crossover wire **58g** is provided as an inter-pole crossover wire **858** positioned in the first slot Sa. An inter-pole crossover wire **759** is arranged at one end of the pseudo tooth **725**, and the inter-pole crossover wire **858** is arranged at the other end of the pseudo tooth **725** opposite to the one end. The inter-pole crossover wire **58g** has a first portion **858g1** arranged at the other end of the pseudo tooth **725**. The first portion **858g1** is located above an end surface of the stator **20** and obliquely arranged in a partial slot Sa1 that is between a single coil **51g** and the pseudo tooth **725**. The first portion **858g1** is provided as a transition portion. The inter-pole crossover wire **58g** has a second portion **858g2** obliquely extending in a space between the pseudo tooth **725** and a single coil **51q**. The second portion **858g2** of the inter-pole crossover wire **58g** extending from the single coil **51g** and a second portion **759p2** the inter-pole crossover wire **59p** extending from the single coil **51q** cross each other in a partial slot Sa2. The second portion **858g2** is provided as a crossed portion.

(157) The second portion **858g2** of the inter-pole crossover wire **58g** extends through an inter-pole gap PG and is located between one end **20a** and the other end **20b** of the stator **20** in the axial direction. The second portion **858g2** extends at least in the circumferential direction and obliquely in the axial direction. As a result, the number of crossover wires **54** stacked in the axial direction or the radial direction on an outer side of the insulator **40** in the radial direction is reduced. The

second portion **759p2** of the inter-pole crossover wire **59p** extends through the inter-pole gap PG and is located between the one end **20a** and the other end **20b** of the stator **20** in the axial direction. The second portion **759p2** extends at least in the circumferential direction and obliquely in the axial direction. As a result, the number of crossover wires **54** stacked in the axial direction or the radial direction on an outer side of an insulator **40** in the radial direction is reduced. The inter-pole crossover wire **58g** and the inter-pole crossover wire **59p** cross each other in the inter-pole gap PG on inner sides of base end flanges **43g** and **43q** in the radial direction. In this embodiment, the inter-pole crossover wire **58g** and the inter-pole crossover wire **59p** cross each other in the partial slot Sa2. Therefore, the crossover wires **54** are shifted from each other in the radial direction within an angle range of the first slot Sa. As a result, the number of crossover wires **54** stacked in the axial direction or the radial direction on an outer side of the insulator **40** in the radial direction is reduced. The number of crossover wires **54** stacked in the axial direction or the radial direction on an outer side of the insulator **40** in the radial direction is limited to two.

(158) FIGS. **33**, **34**, and **35** correspond to FIGS. **27**, **28**, and **29**, respectively. In these drawings, an inter-pole crossover wire **58k** and an inter-pole crossover wire **59g** are arranged at opposite ends of the stator **20** in the axial direction without a pseudo tooth **725**. The inter-pole crossover wire **58k** and the inter-pole crossover wire **59g** are arranged so as to be curved in a space of a second slot Sb. In other words, the inter-pole crossover wire **58k** and the inter-pole crossover wire **59g** do not have a straight shape, i.e., a shortest path. The inter-pole crossover wire **58k** and the inter-pole crossover wire **59g** may have a crank shape.

(159) As shown in FIG. **34**, a second portion **858k2** of the inter-pole crossover wire **58k** extends through an inter-pole gap PG and is located between the one end **20a** and the other end **20b** of the stator **20** in the axial direction. The second portion **858k2** extends at least in the circumferential direction and obliquely in the axial direction. As a result, the number of the crossover wires **54** stacked in the axial direction or the radial direction on the outer side of the insulator **40** in the radial direction is reduced. A second portion **759g2** of the inter-pole crossover wire **59g** extends through the inter-pole gap PG and is located between the one end **20a** and the other end **20b** of the stator **20** in the axial direction. The second portion **759g2** extends at least in the circumferential direction and obliquely in the axial direction. As a result, the number of crossover wires **54** stacked in the axial direction or the radial direction on the outer side of the insulator **40** in the radial direction is reduced. The inter-pole crossover wire **58k** and the inter-pole crossover wire **59g** cross each other in the inter-pole gap PG on an inner side of base end flanges **43k** and **43h** in the radial direction. Therefore, the crossover wires **54** are shifted from each other in the radial direction within an angle range of the second slot Sb. As a result, the number of crossover wires **54** stacked in the axial direction or the radial direction on the outer side of the insulator **40** in the radial direction is reduced. The number of crossover wires **54** stacked in the axial direction or the radial direction on the outer side of the insulator **40** in the radial direction is limited to two.

(160) In this embodiment, the inter-pole crossover wire **58g** and the inter-pole crossover wire **59p** cross each other in the partial slot Sa2. Alternatively, the inter-pole crossover wire **58g** and the inter-pole crossover wire **59p** may cross each other in the partial slot Sa1.

(161) According to the present embodiment, the inter-pole crossover wires **58**, **59** are separately arranged on opposite sides of the pseudo tooth **725** in the axial direction. As a result, advantageous effects of the crossover wires separately arranged can be obtained. For example, physical interaction between multiple crossover wires can be reduced. For example, electrical capacitive components and electrical inductive components can be reduced. Moreover, thermal heat dissipation can be accelerated. Also in the present embodiment, the same effects as those of the preceding embodiments can be obtained.

Ninth Embodiment

(162) This embodiment is a modification of the basic preceding embodiments. In the preceding embodiments, particularly in the seventh embodiment, the winding finish portion **56** of the single

coil **51** is arranged at the tip end portion of the single coil **51**, that is, the inner portion in the radial direction. Alternatively, in the present embodiment, a winding finish portion **56** is arranged at a base end portion of a single coil **51**, that is, at an outer portion in the radial direction.

(163) FIGS. **36**, **37**, and **38** correspond to FIGS. **21**, **23**, and **25**, respectively. In these drawings, the winding finish portions **56g**, **56q** of the single coils **51g**, **51q** are arranged at the base end portions of the single coils **51g**, **51q**. In a first slot Sa, an inter-pole crossover wire **58g** is provided as an inter-pole crossover wire **958** positioned in the first slot Sa. A first portion **958g1** of the inter-pole crossover wire **58g** is obliquely arranged in a partial slot Sa1. The first portion **958g1** is arranged along an inner surface of a yoke **32**. A second portion **958g2** is provided also as a transition portion. The inter-pole crossover wire **58g** and an inter-pole crossover wire **59p** are arranged at an end of a pseudo tooth **725** facing in the axial direction, stacked in the axial direction, and cross each other.

(164) FIGS. **39**, **40**, and **41** correspond to FIGS. **27**, **28**, and **29**, respectively. In these drawings, an inter-pole crossover wire **58k** and an inter-pole crossover wire **59g** are arranged in a second slot Sb without a pseudo tooth **725**. The inter-pole crossover wire **58k** and the inter-pole crossover wire **59g** are stacked in the axial direction and cross each other in a middle portion of an inter-pole gap PG.

(165) According to the present embodiment, the inter-pole crossover wire **58** can be arranged without being limited by the position of the winding finish portion **56**. Similarly, the inter-pole crossover wire **59** may be arranged without being limited by the position of the winding start portion **55**. Also in the present embodiment, the same effects as those of the preceding embodiments can be obtained.

Tenth Embodiment

(166) This embodiment is a modification of the basic preceding embodiments. In the preceding embodiments, the inter-pole crossover wires **58**, **59** are arranged so as to bypass the pseudo teeth **725** or arranged as if the pseudo teeth **725** are virtually presented. Alternatively, the present embodiment includes members that can be replaced with the pseudo teeth **725**.

(167) FIGS. **42**, **43**, and **44** correspond to FIGS. **27**, **28**, and **29**, respectively. FIG. **44** shows a cross section taken along a line XLIV-XLIV of FIG. **42**. In these drawings, an insulator **40** has a protrusion A**46** inside a second slot Sb. The protrusion A**46** contains a resin material. Instead of the pseudo tooth **725**, the protrusion A**46** acts also as a positioning member that fixes the positions the inter-pole crossover wires **58**, **59**. A width of the protrusion A**46** in the circumferential direction is smaller than a width of the second slot Sb in the circumferential direction. Partial slots Sb1 and Sb2 are formed on opposite sides of the protrusion A**46** in the circumferential direction. A length of the protrusions A**46** in the radial direction is shorter than a magnetic pole face of the stator **20**. The protrusion A**46** is arranged in a middle part of the second slot Sb in the circumferential direction. The protrusion A**46** may be displaced from the middle part toward one side of the second slot Sb in the circumferential direction. A height of the protrusion A**46** in the axial direction is approximately $\frac{1}{3}$ of a dimension of the insulator **40** in the axial direction. Alternatively, the height of protrusion A**46** in the axial direction may be equal to the dimension of insulator **40** in the axial direction. The height of the protrusion A**46** in the axial direction can be set so as to obtain strength for fixing the positions of the inter-pole crossover wires **58**, **59**.

(168) An inter-pole crossover wire **58k** is provided as an inter-pole crossover wire **758**. An inter-pole crossover wire **59g** is provided as an inter-pole crossover wire **759**. The inter-pole crossover wires **58k**, **59g** are bent into the illustrated shape by being hooked on the protrusion A**46**. The inter-pole crossover wires **58k**, **59g** are guided by the protrusion A**46** instead of a pseudo tooth **725**. From this point of view, the pseudo teeth **725** and the protrusions A**46** are provided as guide members that guide the inter-pole crossover wires **58**, **59** to their predetermined positions. As a result, inter-pole crossover wires similar to those of the preceding embodiments can be provided.

Eleventh Embodiment

(169) This embodiment is a modification of the basic preceding embodiments. In the preceding

embodiments, the pseudo teeth **725** and the protrusions **A46** are provided as positioning members on which the inter-pole crossover wires **58**, **59** are just hooked. Alternatively, the present embodiment introduces grooves to more securely hold inter-pole crossover wires **58,59**.

(170) FIGS. **45**, **46**, and **47** correspond to FIGS. **27**, **28**, and **29**, respectively. FIG. **47** shows a cross section taken along a line XLVII-XLVII in FIG. **45**. In these drawings, an insulator **40** has a protrusion **B46** inside a second slot **Sb**. The protrusion **B46** has approximately the same size as the protrusion **A46** of the preceding embodiment. The protrusion **B46** has a groove **B47** that receives the inter-pole crossover wire **59g**. The protrusion **B46** has a groove **B48** that receives the inter-pole crossover wire **58k**. The groove **B47** is arranged on one end surface of the protrusion **B46** in the axial direction. The groove **B48** is arranged on another end surface of the protrusion **B46** in the axial direction. The groove **B47** and the groove **B48** are arranged on the opposite surfaces of the protrusion **B46**. The grooves **B47**, **B48** hold the inter-pole crossover wires **59g**, **58k**.

(171) The groove **B47** restricts movement of the inter-pole crossover wire **59g** at least in the radial direction. The groove **B47** is a relatively small groove that receives the inter-pole crossover wire **59g** while restricting its movement. The inter-pole crossover wire **59g** is press-fitted into the groove **B47** in the axial direction. The groove **B47** has a longitudinal direction in the circumferential direction or in a tangential direction. The groove **B47** is also called a snap-fit groove that restricts movement of the inter-pole crossover wire **59g** by elasticity of the protrusion **B46**. As a result, the groove **B47** restricts the movement of the inter-pole crossover wire **59g** also in the circumferential direction or the tangential direction. A depth of the groove **B47** in the axial direction is equal to or greater than a radius of the inter-pole crossover wire **59g**. The groove **B47** may be a relatively large groove that loosely receives the inter-pole crossover wire **59g** in a floatable state. The depth of the groove **B47** in the axial direction may be less than the radius of the inter-pole crossover wire **59g**.

(172) The groove **B48** restricts movement of the inter-pole crossover wire **58k** at least in the radial direction. The groove **B48** is a relatively small groove that receives the inter-pole crossover wire **58k** while restricting its movement. The inter-pole crossover wire **58k** is press-fitted into the groove **B48** in the axial direction. The groove **B48** has a longitudinal direction in the circumferential direction or in the tangential direction. The groove **B48** is also called a snap-fit groove that restricts movement of the inter-pole crossover wire **58k** by elasticity of the protrusion **B46**. As a result, the groove **B48** restricts the movement of the inter-pole crossover wire **58k** also in the circumferential direction or the tangential direction. A depth of the groove **B48** in the axial direction is equal to or greater than a radius of the inter-pole crossover wire **58k**. The groove **B48** may be a relatively large groove that loosely receives the inter-pole crossover wire **58k** in a floatable state. The depth of the groove **B48** in the axial direction may be less than the radius of the inter-pole crossover wire **58k**.

(173) In a stator manufacturing method, restraining and holding the inter-pole crossover wire arranged on the stator contribute to improvement in stability of the shape of the inter-pole crossover wire at a winding step. At the winding step, after one single coil **51k** is wound, the inter-pole crossover wire **58k** is accommodated and held in the groove **B48**. This prevents the winding of the single coil **51** from loosening loosely. At the winding step, after the end crossover wire **57g** is arranged, the inter-pole crossover wire **59g** is accommodated and held in the groove **B47**. This prevents the end crossover wire **57** from loosening loosely. As a result, the groove **B47** and the groove **B48** improve the shape stability of the coil **50**. In addition, the groove **B47** and the groove **B48** promote the process at the winding step.

(174) An inter-pole crossover wire **58k** is provided as an inter-pole crossover wire **B58**. An intermediate portion of the inter-pole crossover wire **B58** is restrained by the protrusion **B46**. As a result, the inter-pole crossover wire **B58** is arranged to be curved in a Z-shape or an S-shape. An inter-pole crossover wire **59g** is provided as an inter-pole crossover wire **B59**. An intermediate portion of the inter-pole crossover wire **B59** is restrained by the protrusion **B46**. As a result, the inter-pole crossover wire **B59** is arranged to be curved in a Z-shape or an S-shape.

(175) In the present embodiment, the stator **20** includes both the grooves **B47** and **B48**.

Alternatively, the stator **20** may have only one of the grooves **B47** and **B48**. In the present embodiment, a groove **B47** or a groove **B48** is provided on a resin protrusion **B46** extending from the insulator **40**. Alternatively, a groove **B47** or a groove **B48** may be provided on insulators **740a** and **740b** of the pseudo teeth **725**. Also in the present embodiment, the same effects as those of the preceding embodiments can be obtained.

Twelfth Embodiment

(176) This embodiment is a modification of the basic preceding embodiments. As shown in the preceding embodiments, also in the eleventh embodiment, the position of the winding finish portion **56** of the single coil **51** can be changed variously.

(177) FIGS. **48**, **49**, and **50** correspond to FIGS. **27**, **28**, and **29**, respectively. FIG. **50** shows a cross section taken along a line L-L in FIG. **48**. In these drawings, a winding finish portion **56k** of a single coil **51k** is positioned at an outer portion in the radial direction. An inter-pole crossover wire **58k** is provided as an inter-pole crossover wire **C58**. The inter-pole crossover wire **C58** is arranged along an inner surface of a stator core **30**. Also in the present embodiment, the same effects as those of the preceding embodiments can be obtained.

Thirteenth Embodiment

(178) This embodiment is a modification of the basic preceding embodiments. In the preceding embodiments, the rotor magnetic poles **12** are arranged at equal intervals. Alternatively, some of the magnetic poles included in the rotor magnetic poles **12** may be arranged slightly shifted from the equidistant positions. A rotor **10** disclosed in the present embodiment can be used as the rotors of the preceding embodiments.

(179) In FIG. **51**, most of the rotor magnetic poles **12** are arranged at equal intervals. For example, three magnetic poles **12a**, **12b**, **12c** adjacent to each other are separated from each other by an interval **G1** ($G1=G1$). The rotor magnetic poles **12** includes one or more rotor magnetic poles **12** that are displaced in the circumferential direction. The one or more rotor magnetic poles **12** displaced in the circumferential direction are called displaced magnetic poles. For example, two magnetic poles **12d** and **12e** adjacent to each other are separated by an interval **G2**. Two magnetic poles **12e** and **12f** adjacent to each other are separated by an interval **G3**. The interval **G2** and the interval **G3** are not equal ($G2 \neq G3$). The relationship between the interval **G2** and the interval **G3** is $G2 < G3$ or $G2 > G3$. As a result, the magnetic pole **12e** is shifted in an advancing direction or in a retarding direction. The magnetic pole **12e** is provided as a displaced magnetic pole. An amount of displacement is a very small amount that is difficult to be shown in the drawings. The rotor **10** may have a single displaced magnetic pole or multiple displaced magnetic poles. For example, the rotor **10** may have three displaced magnetic poles. The displaced magnetic pole is capable of adjusting a torque fluctuation observed in a rotor **10** to have a desired waveform.

(180) The disclosure in this specification, drawings and the like is not limited to the exemplified embodiments. The disclosure encompasses the illustrated embodiments and variations based on the embodiments by those skilled in the art. For example, the present disclosure is not limited to the combinations of components and/or elements shown in the embodiments. The present disclosure may be implemented in various combinations. The present disclosure may have additional members which may be added to the embodiments. The present disclosure encompasses the embodiments where some components and/or elements are omitted. The present disclosure encompasses replacement or combination of components and/or elements between one embodiment and another. The disclosed technical scope is not limited to the descriptions of the embodiments.

(181) In the above embodiments, the rotary electric machine **1** is an inner rotor type. Alternatively, the rotary electric machine **1** may be an outer rotor type. In the above embodiments, the rotary electric machine **1** provides an electric motor. Alternatively, the rotary electric machine **1** may provide a generator or a generator motor. Further, the rotary electric machine **1** can be used for various purposes such as a servo motor and a step motor.

(182) In the above embodiments, the stator core **30** is provided by a steel plate in which the multiple teeth **31** and the yoke **32** are continuous. Alternatively, the stator core **30** may be provided by a so-called multi-split core. In this case, the stator core **30** is provided by a connected body of multiple partial cores. For example, one partial core is provided by a continuous body of one partial-annular partial yoke and one tooth.

(183) In the above embodiments, the multiple conductive members **60** are insert-molded on the terminal blocks **80**, **280**. Alternatively, the multiple conductive members **60** may be press-fitted and fixed to the terminal blocks **80**, **280**. Further, the multiple conductive members **60** may be supported by or fixed to the insulator **40** without the terminal blocks **80**, **280**. For example, the multiple conductive members **60** may be directly fixed to the insulator **40** by snap fitting. Also in this configuration, since the connection portions **65**, **66**, **67**, **68**, **69**, **270** are arranged in the inter-pole gaps PG, the physical size of the coil end **52** of the rotary electric machine **1** can be reduced. In the above embodiments, the connector portion **82** has an opening facing outward in the radial direction in the outside of the housing **3**. Alternatively, the connector portion **82** may have an opening facing in the axial direction outside the housing **3**. In this case, the connector portion **82** forms an electrical connection by receiving a connector of the external circuit moved along either of the axial directions.

(184) In the above embodiments, the coil **50** is provided by star connection. Alternatively, the coil **50** may be provided by delta connection. In this case, the connection portion between one conductive member **60** and at least two coil ends **52** is arranged in the inter-pole gap PG. Further, in the above embodiments, one phase winding is provided by one wire. Alternatively, one phase winding may be provided by multiple wires as a parallel circuit. In this case, one single coil **51** is provided by the parallel circuit of the multiple wires. For example, when one single coil **51** is provided by two wires, a connection portion between the conductive member **60** that provides one power terminal and two coil ends **52**, in the star connection, is arranged in the inter-pole gap PG. For example, when one single coil **51** is provided by two wires, a connection portion between the conductive member **60** that provides one power terminal and four coil ends **52**, in the delta connection, is arranged in the inter-pole gap PG.

(185) In the above embodiments, the conductive member **60** is a bus bar. Alternatively, the conductive member **60** may be an electrode for a terminal, a lead wire, or a conductor foil on a substrate. Also in these cases, since the connection portion is arranged in the inter-pole gap PG, miniaturization can be achieved. In the above embodiments, the coil **50** is made of copper or a copper alloy. Alternatively, the coil **50** may be made of aluminum or aluminum alloy. In the above embodiments, the connection between the conductive member and the coil end **52** is provided by fusing. Alternatively, the connection between the conductive member and the coil end **52** may be provided by heat staking, welding, soldering, for example.

(186) In the above embodiments, the conductive member **60** has the connector terminal extending in the radial direction as the power end. Alternatively, the conductive member **60** may extend along the axial direction. Also in this cases, since the connection portion is arranged in the inter-pole gap PG, miniaturization in the axial direction can be achieved. In the above embodiments, the multiple power end members **61**, **62**, **63** provide connector terminals. Alternatively, the power end members **61**, **62**, **63** may provide press-fit terminals, solder terminals, for example.

(187) In the above embodiments, the inter-pole gap PG is a cavity. Alternatively, the inter-pole gap PG may be filled with a resin member after disposing the conductive member **60**. Further, the conductive member **60** arranged in the inter-pole gap PG may be coated with a thin resin material. In any configuration, the connection portion which is a part of the conductive member **60** is disposed in the inter-pole gap PG. Thus, the miniaturized rotary electric machine **1** is provided.

(188) In the above embodiments, the multiple coil ends at winding start are used the power ends, and the multiple coil ends at winding end are used as the neutral points. Alternatively, the multiple coil ends at winding end may be used the power ends, and the multiple coil ends at winding start

may be used as the neutral points. Further, the coil end may be the power end or the neutral point without using the conductive member **60**. For example, the multiple coil ends at the start or end of winding may be drawn out long as power ends. For example, the multiple coil ends at the start or end of winding may be directly joined as neutral points.

(189) While the present disclosure has been described with reference to embodiments thereof, it is to be understood that the disclosure is not limited to the embodiments and constructions. To the contrary, the present disclosure is intended to cover various modification and equivalent arrangements. In addition, while the various elements are shown in various combinations and configurations, which are exemplary, other combinations and configurations, including more, less or only a single element, are also within the spirit and scope of the present disclosure.

Claims

1. A rotary electric machine, comprising: a multi-phase winding including magnetic poles circumferentially arranged in a circumferential direction, the magnetic poles protruding inward from an annular yoke in a radial direction; and a positioning member positioned between two of the magnetic poles that are adjacent to each other in the circumferential direction, the positioning member protruding inward from the annular yoke in the radial direction, wherein: the multi-phase winding includes crossover wires; each crossover wire connects two single coils functioning as two of the magnetic poles that are same in phase; the crossover wires include: an inter-pole crossover wire extending between one end and another end of the multi-phase winding in an axial direction of the multi-phase winding, the inter-pole crossover wire extending through an inter-pole gap between the adjacent two of the magnetic poles; and an end crossover wire extending in the circumferential direction on the other end of the multi-phase winding; and the inter-pole crossover wire is arranged to be capable of bypassing the positioning member in the inter-pole gap.
2. The rotary electric machine according to claim 1, wherein the positioning member is a pseudo tooth made of a magnetic material.
3. The rotary electric machine according to claim 1, wherein the positioning member is a protrusion made of a resin material.
4. The rotary electric machine according to claim 1, wherein the positioning member includes a groove that holds the inter-pole crossover wire.
5. The rotary electric machine according to claim 1, wherein: the inter-pole gap is one of a plurality of inter-pole gaps of a stator; the plurality of inter-pole gaps include first inter-pole gaps in each of which the positioning member is provided, and second inter-pole gaps in each of which the positioning member is not provided; the inter-pole crossover wire bypasses the positioning member in each of the first inter-pole gaps; and the inter-pole crossover wire bypasses a positioning member virtually presented in each of the second inter-pole gaps.
6. The rotary electric machine according to claim 1, wherein the crossover wires include end crossover wires which are different in phase and are arranged in the axial direction on the other end of the multi-phase winding, and a number of the end crossover wires arranged in the axial direction is two or less at every position in the circumferential direction.
7. A rotary electric machine, comprising: a multi-phase winding including magnetic poles circumferentially arranged in a circumferential direction; and a positioning member positioned between two of the magnetic poles that are adjacent to each other in the circumferential direction, wherein: the multi-phase winding includes crossover wires; each crossover wire connects two single coils functioning as two of the magnetic poles that are same in phase; the crossover wires include: an inter-pole crossover wire extending between one end and another end of the multi-phase winding in an axial direction of the multi-phase winding, the inter-pole crossover wire extending through an inter-pole gap between the adjacent two of the magnetic poles; and an end crossover wire extending in the circumferential direction on the other end of the multi-phase

winding; the inter-pole crossover wire bypasses the positioning member; and the positioning member is a pseudo tooth made of a magnetic material.

8. A rotary electric machine, comprising a multi-phase winding including magnetic poles circumferentially arranged in a circumferential direction, wherein: the multi-phase winding includes crossover wires; each crossover wire connects two single coils functioning as two of the magnetic poles that are same in phase; the crossover wires include: an inter-pole crossover wire extending between one end and another end of the multi-phase winding in an axial direction of the multi-phase winding, the inter-pole crossover wire extending through an inter-pole gap between an adjacent two of the magnetic poles; and an end crossover wire extending in the circumferential direction on the other end of the multi-phase winding; the inter-pole crossover wire bypasses a positioning member in the inter-pole gap; the crossover wires include end crossover wires which are different in phase and are arranged in the axial direction on the other end of the multi-phase winding; and a number of the end crossover wires arranged in the axial direction is two or less at every position in the circumferential direction.
